

Homework 09: NumPY and Matplotlib

EEC 3150

For each Python exercise, write a separate Python script. Submit all scripts on Blackboard under the appropriate assignment. Name your script files with the following format:

HW<assignment number>_Ex<exercise
number>_<FirstInitial><LastName>.py

Example: HW01_Ex01_GWest.py

For short answer or math exercises, either do your work on a sheet of paper and upload a picture of it to Blackboard or submit some kind of text file or Word document to Blackboard. You can place all such exercises in a single document as long as you clearly label each exercise. These files should follow a similar naming convention as the Python scripts.

Exercise 1 (*10 points, Python*)

2D random walk, unit circle displacement

Exercise 2 (*10 points, Python*)

Create a 1D random walk (with your choice of distribution) which will ALWAYS stay within some bounded domain of values
plot the walk as a blue line plot the upper and lower bounds as a dotted red line

Exercise 3 (*10 points, Python*)

randomly generate points on the surface of a cylinder

Exercise 4 (*10 points, Python*)

use MC methods to integrate $\sin(x)$ from $0-3\pi$
(figure out how to handle negative values)